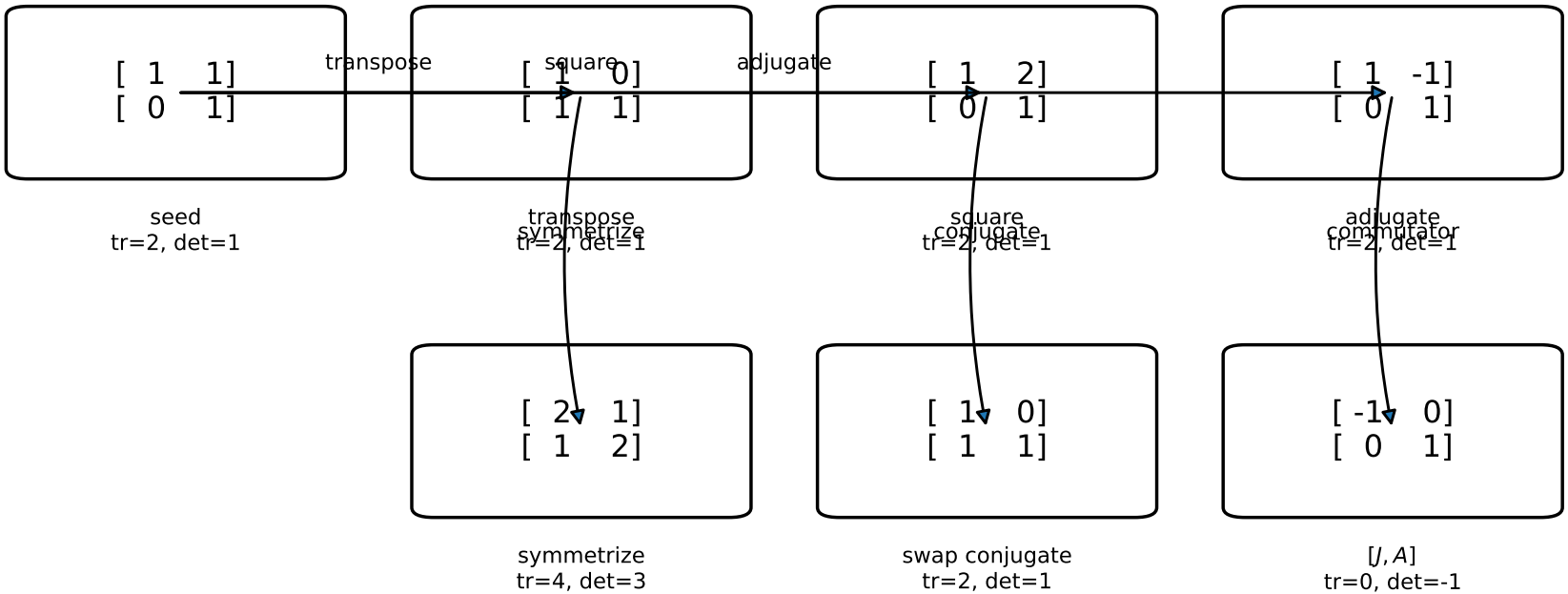Matrix deformation atlas with exact trace and determinant metadata